Module 5 Week A — Stretch: Learning Curves Diagnostic

Honors Track. This is an Honors Track stretch assignment. Stretch assignments are for learners who have completed all core assignments, are On Track or Advanced, and are attending consistently. If you are behind on core work, focus there first.

Challenge tier prerequisite: None – this stretch builds on this week's base integration task.

The Challenge

Implement `learning_curve` from scikit-learn to diagnose whether the logistic regression model on the telecom churn dataset is suffering from high bias or high variance. Produce a learning curve plot (training set size versus score for both training and validation sets), then write a short analysis connecting the curves to the bias-variance tradeoff concepts from the reading.

Learning curves are one of the most useful diagnostic tools in applied ML. They answer practical questions that metrics alone cannot: Would collecting more data help? Would a more complex model help? Is the model fundamentally limited by its capacity? Your analysis should address these questions directly.

What You'll Learn

- Using `sklearn.model_selection.learning_curve` to generate training and validation scores across dataset sizes
- Interpreting the gap between training and validation curves as a bias-variance diagnostic
- Connecting learning curve shapes to actionable model improvement strategies
- Communicating diagnostic findings in a concise written analysis

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Outcome

Your deliverable is a standalone repository containing:

Artifact	Description
Python script or notebook	Uses `learning_curve` to compute training and validation scores, then generates the plot
Learning curve plot	Training set size on the x-axis, score on the y-axis, with separate lines for training and validation performance (include confidence bands)
Written analysis (1–2 paragraphs)	Interprets the learning curves and answers the diagnostic questions below

Constraints:

- Use `sklearn.model_selection.learning_curve` with at least 5 different training set sizes
- Use stratified cross-validation (the dataset has class imbalance)
- Plot both training and validation scores with shaded regions showing +/- 1 standard deviation across folds
- Choose an appropriate scoring metric given the class imbalance in the telecom dataset (accuracy may not be the best choice – justify your selection)

Your written analysis must address:

1. Based on the learning curve shape, is the model suffering primarily from high bias (underfitting) or high variance (overfitting)?
2. Would collecting more data likely improve validation performance? What evidence from the curves supports your answer?
3. Would increasing model complexity (e.g., adding polynomial features, switching to a more flexible model) likely help? Why or why not?
4. What is your recommended next step for improving this model?

Tips

- A model with high bias shows both training and validation scores converging at a low value – more data will not help because the model is too simple.
- A model with high variance shows a large gap between training and validation scores – more data may help because the model is memorizing rather than generalizing.
- The standard deviation bands are important. If the validation score confidence interval is wide, the model's performance is unstable across folds – this is itself a diagnostic signal.
- Consider generating learning curves for more than one model (e.g., logistic regression with different regularization strengths, or logistic regression versus a more complex model) to make the bias-variance tradeoff concrete.

Submission

1. Create a standalone repository for this stretch assignment (you may use the same repo as Stretch 5A-S1 if you completed it, or create a new one)
2. Push your script, plot, and written analysis to the repository
3. Paste your repository URL into TalentLMS -> Module 5 -> Stretch 5A-S2